

Week 1: Linux Fundamentals

Machine Test: Day3 (Linux file operations and editors)

Name: Danish John

Batch: Horizon

Commands used:

- mkdir
- ls
- cd
- touch
- cp
- mv
- nano
- echo
- cat
- head
- tail
- wc
- wc -l
- file
- vim

Task 1 – File Creation

1. Create a folder named linux-test.
2. Create the following files: devops.txt, cloud.txt, aws.txt.
3. Verify all files are created.

```
xyaro@void:~/devops$ mkdir linux-test
xyaro@void:~/devops$ touch devops.txt cloud.txt aws.txt
xyaro@void:~/devops$ ls
aws.txt  cloud.txt  devops.txt  linux-test
xyaro@void:~/devops$
```

Task 2 – File Copy Operations

1. Create a folder named backup.
2. Copy devops.txt into the backup folder.
3. Copy both cloud.txt and aws.txt into the backup folder.
4. Verify the copied files.

```
xyaro@void:~/devops$ mkdir backup
xyaro@void:~/devops$ cp devops.txt backup
xyaro@void:~/devops$ cp cloud.txt aws.txt backup
xyaro@void:~/devops$ ls
aws.txt  backup  cloud.txt  devops.txt  linux-test
xyaro@void:~/devops$ cd backup/
xyaro@void:~/devops/backup$ ls
aws.txt  cloud.txt  devops.txt
xyaro@void:~/devops/backup$
xyaro@void:~/devops/backup$ |
```

Task 3 – File Rename & Move

1. Rename cloud.txt to linux.txt.
2. Create a folder named documents.
3. Move linux.txt into the documents folder.
4. Verify the result.

```
xyaro@void:~/devops$ mv cloud.txt linux.txt
xyaro@void:~/devops$ mkdir documents
xyaro@void:~/devops$ mv linux.txt documents/
xyaro@void:~/devops$ ls
aws.txt  backup  devops.txt  documents  linux-test
xyaro@void:~/devops$ cd documents/
xyaro@void:~/devops/documents$ ls
linux.txt
xyaro@void:~/devops/documents$ |
```

Task 4 – File Content Management

1. Add the following content to devops.txt
Linux Fundamentals
Devops Engineering
Cloud Computing
2. Display the file content
3. Append a new line: Learning Linux Commands
4. Display the updated content

```
xyaro@void:~/devops$ touch notes.txt
xyaro@void:~/devops$ nano notes.txt
xyaro@void:~/devops$ cat notes.txt
Linux Administration, DevOps Practices
xyaro@void:~/devops$ |
```

Task 5 – Nano Editor

1. Create a file named notes.txt.
2. Open it using Nano.
3. Add: Linux Administration, DevOps Practices.
4. Save and exit.
5. Display the content.

```
xyaro@void:~/devops$ nano devops.txt
xyaro@void:~/devops$ cat devops.txt
Linux Fundamentals
Devops Engineering
Cloud Computing
xyaro@void:~/devops$ echo "Learning Linux Commands" >> devops.txt
xyaro@void:~/devops$ cat devops.txt
Linux Fundamentals
Devops Engineering
Cloud Computing
Learning Linux Commands
xyaro@void:~/devops$ |
```

Task 6 – File Viewing Commands

Using notes.txt: Display the full content, open the file using a command suitable for reading large files, and exit properly.

```
xyaro@void:~/devops$ less notes.txt
xyaro@void:~/devops$ |
```

```
Linux Administration, DevOps Practices
notes.txt (END)
```

Task 7 – Head & Tail

1. Create logs.txt.
2. Add at least 15 lines of text.
3. Display the first 10 lines and last 10 lines.

```
xyaro@void:~/devops$ touch logs.txt
xyaro@void:~/devops$ nano logs.txt
xyaro@void:~/devops$ head logs.txt
1 [09:00:00] [SYSTEM] Starting up all core computer services...
2 [09:00:02] [SUCCESS] Hard drive and memory check completed with zero errors.
3 [09:00:05] [INFO] Connecting to the local Wi-Fi network...
4 [09:00:07] [SUCCESS] Internet connection established successfully.
5 [09:01:15] [INFO] User 'Alex' logged into the desktop dashboard.
6 [09:02:40] [INFO] Opening Google Chrome browser.
7 [09:05:11] [INFO] Opening Spotify app to play music.
8 [09:15:22] [WARNING] Battery level has dropped below 20%. Please plug in.
9 [09:15:30] [INFO] Power cable plugged in. Device is now charging.
10 [09:22:04] [INFO] Automatic system update check started.
xyaro@void:~/devops$ tail logs.txt
6 [09:02:40] [INFO] Opening Google Chrome browser.
7 [09:05:11] [INFO] Opening Spotify app to play music.
8 [09:15:22] [WARNING] Battery level has dropped below 20%. Please plug in.
9 [09:15:30] [INFO] Power cable plugged in. Device is now charging.
10 [09:22:04] [INFO] Automatic system update check started.
11 [09:22:15] [INFO] Downloading a new security update (File size: 45 MB).
12 [09:23:00] [SUCCESS] Security update installed without any issues.
13 [09:30:45] [INFO] User 'Alex' clicked 'Shut Down Device'.
14 [09:30:48] [INFO] Closing all open programs and saving data...
15 [09:31:00] [SYSTEM] Turn off complete. System is now
xyaro@void:~/devops$ |
```

Task 8 – Word Count

Using logs.txt: Display the number of lines, words, and characters. Display only the line count.

```
xyaro@void:~/devops$ wc logs.txt
 15 143 978 logs.txt
xyaro@void:~/devops$ wc -l logs.txt
15 logs.txt
xyaro@void:~/devops$ |
```

Task 9 – File Type Identification

Check the type of devops.txt and logs.txt and record the output.

```
xyaro@void:~/devops$ file devops.txt
devops.txt: ASCII text
xyaro@void:~/devops$ file logs.txt
logs.txt: ASCII text
xyaro@void:~/devops$ |
```

Task 10 – Vim Editor

Create config.txt, open using Vim, add PORT=3000 and ENV=production, save and exit, then display the content.

```
xyaro@void:~/devops$ touch config.txt
xyaro@void:~/devops$ vim config.txt
xyaro@void:~/devops$ cat config.txt
PORT=3000
ENV=production
xyaro@void:~/devops$ |
```

Interview Questions

1. What is the purpose of the touch command?
touch command is used to create an empty file.
2. Difference between cp and mv?
cp is using for copy files, directories, copy content of a file.
mv is using for rename the files, directories and move a file or directory one place to another.
3. How do you rename a file in Linux?
Using **mv** command [syntax -> **mv filename new_filename**]
4. Difference between > and >>?
> is for add content into a file.
>> is for append the content into a file.
5. What is the purpose of the cat command?
It is using for display the file content in the terminal.
6. Difference between cat and less?
cat: display the content of a file in the terminal.
less: display the content of a large file with scrollable feature.
7. Why is less preferred for large files?
Because it helps users to scroll and view the large contents and it doesn't load the terminal screen.

8. What does the head command do?
It is using for display the first 10 lines of content in a file.
9. What does the tail command do?
It is using for display the last 10 lines of content in a file.
10. What is the purpose of tail -f?
It is using for display the last 10 lines of content and it is live, so when the lines add it display that new line live in the bottom of the lines in the terminal.
11. What information does wc provide?
Lines, words, characters in a file.
12. What does wc -l count?
It counts the total lines in a file.
13. What is the purpose of the file command?
To determine the actual type of a file.
14. Difference between Nano and Vim?
Nano is a beginner friendly text editor
Vim is a advanced text editor tool
15. How do you save and exit Vim?
By typing `:wq` and press enter in normal mode.
16. How do you exit Vim without saving?
By typing `:q!` and press enter.
17. Why is Vim important for DevOps engineers?
Vim is an essential tool for DevOps engineers because they spend most of their time configuring, managing, and troubleshooting remote servers where graphical user interfaces (GUIs) do not exist.
18. What is the purpose of cp -r?
`cp -r` is using for copy the directory with it's contents (files and directories of the directory).
19. Which command is used to monitor logs in real time?
`tail -f <filename>`
20. Which command is used to identify a file type?
`file <filename>`

Final Folder Structure Output

```
xyaro@void:~/devops$ tree
```

```
.
├── aws.txt
├── backup
│   ├── aws.txt
│   ├── cloud.txt
│   └── devops.txt
├── config.txt
├── devops.txt
├── documents
│   └── linux.txt
├── linux-test
├── logs.txt
└── notes.txt
```

```
4 directories, 9 files
```

```
xyaro@void:~/devops$ |
```